

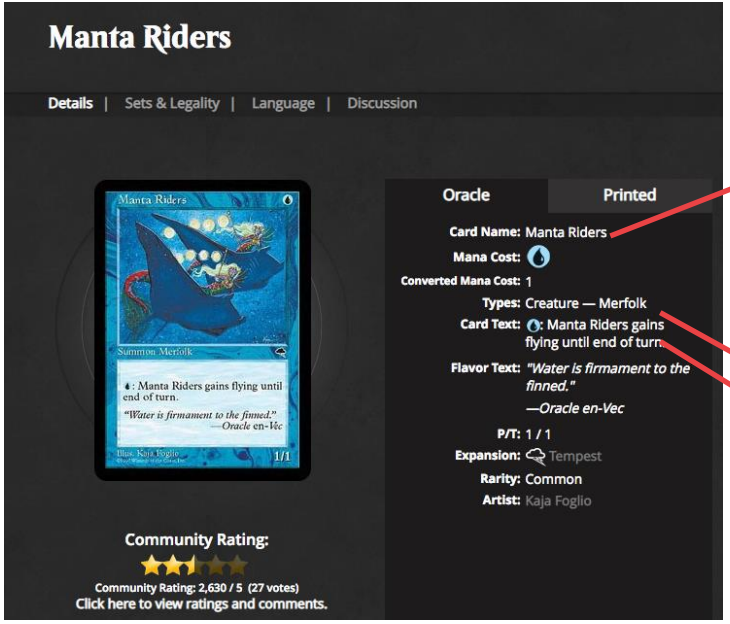
Create MTG Trading Card Database By Crawler

Practical Exam

Goal

magicthegathering.io provides up-to-date information regarding all MTG trading cards available:

- https://gatherer.wizards.com/




```
<div class="smallGreyMono" style="margin-top: 5px;">
  <b class="ft"><b></b></b>
  <div id="ctl00_ctl00_ctl00_MainContent_SubContent_SubContent_nameRow" class="row">
    <div class="label">
      Card Name:</div>
    <div class="value">
      Manta Riders</div>
  </div>
  <div id="ctl00_ctl00_ctl00_MainContent_SubContent_SubContent_manaRow" class="row manaRow">
    <div class="label" style="line-height: 25px;">
      Mana Cost:</div>
    <div class="value">
      <img src="/Handlers/Image.ashx?size=medium&amp;name=U&amp;type=symbol" alt="Blue" align="absbottom" /></div>
  </div>
  <div id="ctl00_ctl00_ctl00_MainContent_SubContent_SubContent_cmcRow" class="row">
    <div class="label" style="font-size: .7em;">
      Converted Mana Cost:</div>
    <div class="value">
      1</div>
  </div>
  <div id="ctl00_ctl00_ctl00_MainContent_SubContent_SubContent_typeRow" class="row">
    <div class="label">
      Types:</div>
    <div class="value">
      Creature — Merfolk</div>
  </div>
  <div id="ctl00_ctl00_ctl00_MainContent_SubContent_SubContent_textRow" class="row">
    <div class="label">
      Card Text:</div>
    <div class="value">
      <div class="cardtextbox" style="padding-left:10px;"><img src="/Handlers/Image.ashx?size=small&amp;name=U&amp;type=symbol" alt="Blue" align="absbottom" />: Manta Riders gains flying until end of turn.</div></div>
  </div>
```

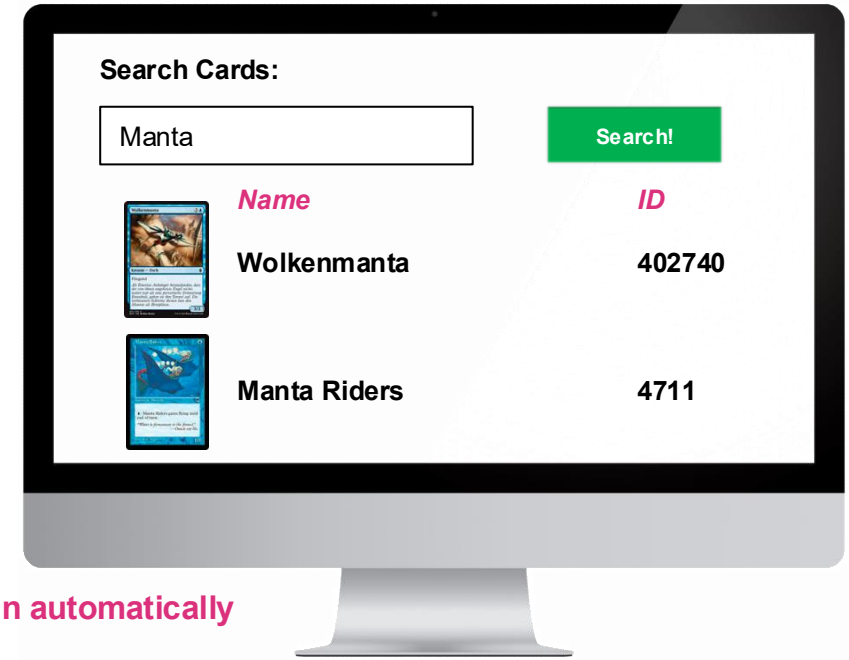
http://gatherer.wizards.com/Pages/Card/Details.aspx?multiverseid=4711

Goal

We want to make use of this data to build a searchable database of all MTG trading cards.

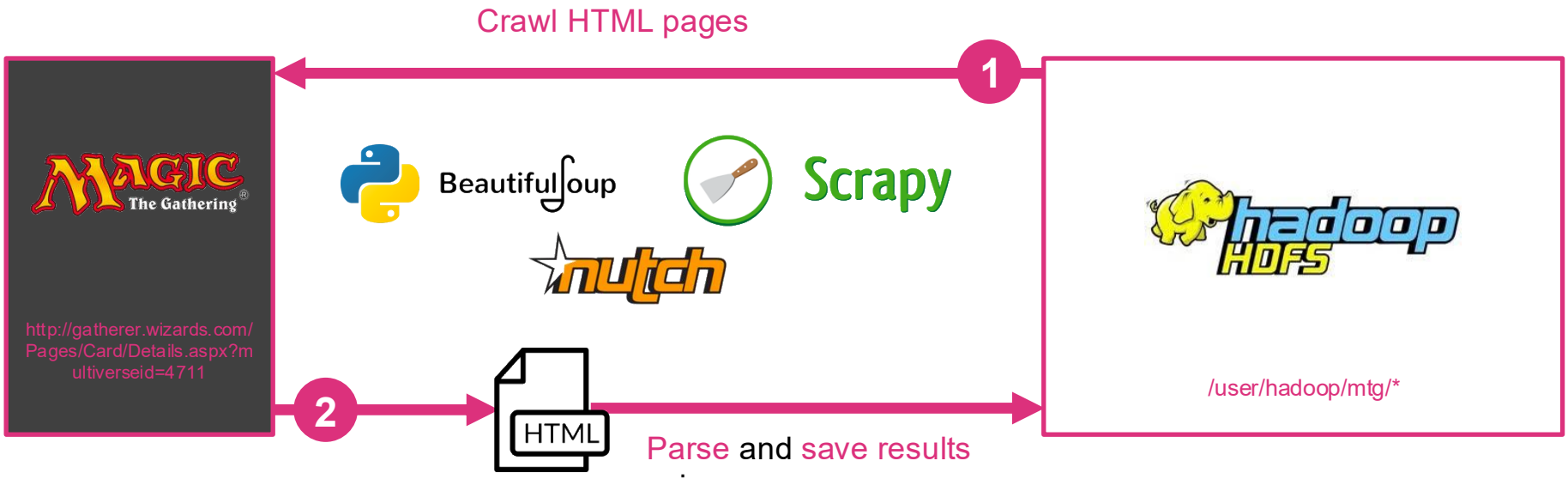
Workflow:

- **Crawl data** from gatherer.wizards.com
- **Parse** required **information** and save them to **HDFS** (queryable through Hive)
- **Export** MTG data **to end-user database** (e.g. MySQL, MongoDB…)
- Provide a simple **HTML Frontend** which is able to:
  - read from end-user database
  - process user input (card name, text or artist)
  - **display search results**



- The whole data workflow **must be implemented** within an ETL **workflow tool** (e.g. Pentaho Data Integration or Airflow) and **run automatically**

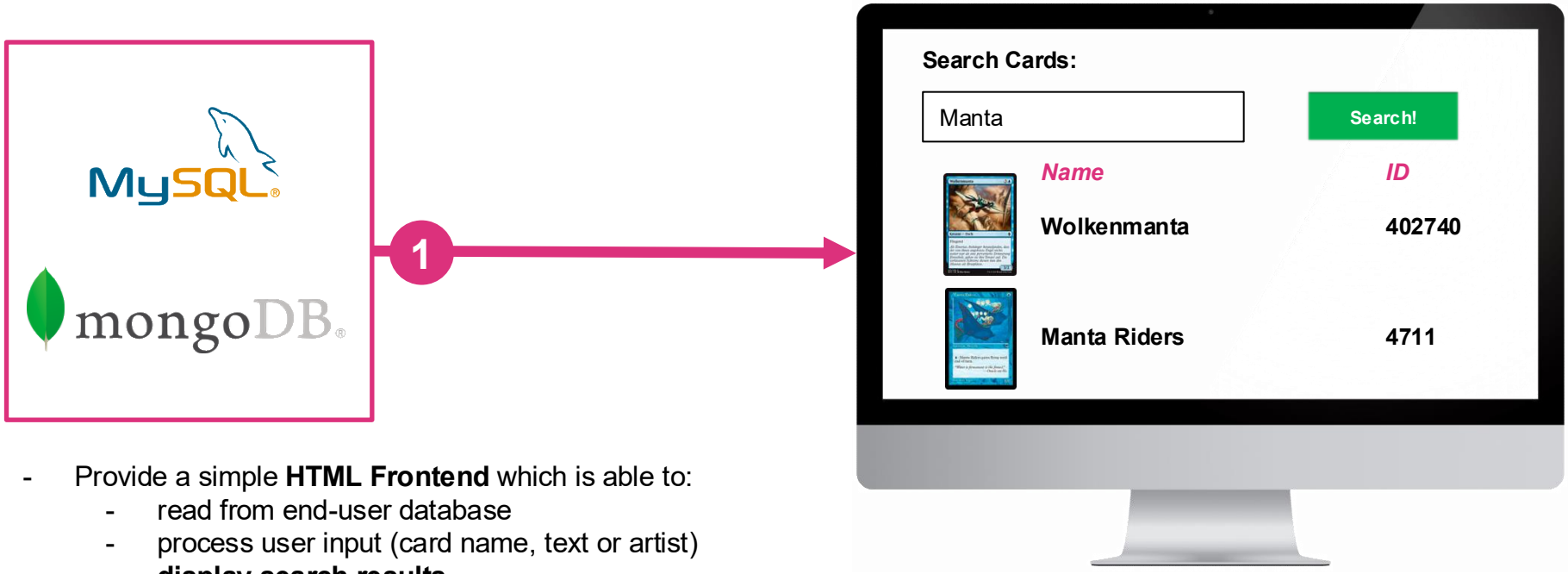
Dataflow: 1. Get MTG Data



Dataflow: 3. Enhance Data And Save Results



Dataflow: 4. Provide Simple Web Interface



- Provide a simple **HTML Frontend** which is able to:
  - read from end-user database
  - process user input (card name, text or artist)
  - **display search results**